

Final Presentations

To demonstrate the project that you did, you are asked to present a presentation that captures your work of the last weeks.

We are not aiming to evaluate you on your presenting and graphic design skills, but instead on the project that you did. The presentations are there to help us understand what you did in your project.

We recommend some topics that you should put in your presentation. You may deviate from this as necessary, but please consider that these specific topics will help us with grading.

The presentation should be 10 minutes, followed by 2.5 minutes for questions. You will be presenting to your TA and Ivo. Keep in mind that Ivo does not know the details of your project.

Please submit the presentation and a .zip of your code on June 13th end of day. Also make sure your TA has access to your GitHub repo. This assignment accounts for 50% of your final grade and needs to be at least a 5.0 to pass the course.

Your presentation is expected to reflect several items that we need to know to assess the quality of your project. These are (elaborated on the next page):

1. How your model can be used to address a problem (introduction)
 Optionally: what you wanted to find out with your project
2. Why your project has novelty
3. How you preprocessed your data
 Highlight: how you train/test/validate split
4. Which model(s) you chose to implement
 - a. Include hyperparameter tuning and regularization
5. How your model(s) performed (results)
 - a. Evaluate whether there is overfitting
 - b. Use relevant metrics
 - c. Compare your model against the baseline properly
6. What you did to make your project professional
7. Which extras you did that may boost your grade
8. Insights you gained from the project as related to the problem at hand

Note that you do not need to only show “good” outcomes. Sometimes failed models and pipelines can also be insightful.

Aside from these points you will also be specifically assessed on your project quality (any issues that it may have) and on the insight your project attempted to generate or generated.

1. How your model can be used to address a problem (Introduction)

This point aims to have you demonstrate what the task is beyond the Machine Learning. It should explain the context in which your model should be deployed. What might happen before your API receives an input, and what might happen with the output that your API gives. If there was anything you wanted to find out about the problem through your project you could also put that here.

Example: A user records an audio fragment of bird-song. Then the user should select roughly which part of the recording contains the bird song. They then send this to the API which cuts the leading and trailing silences, extracts features and delivers the most likely predictions. The user can then interpret this. Previous methods have used time-frequency based features, but we're wondering if we can reach adequate model performance considering only frequency based features.

2. Why your project has novelty

You should address what's interesting/special about your specific project. You've argued this before in your project proposal, but of course your project may have shifted. At this point you need to briefly pitch why your project has some amount of novelty.

3. How you preprocessed your data (Methods)

What preprocessing steps did you do? Why did you choose these? If you experimented with various preprocessing pipelines you are encouraged to discuss those experiments. It should be sufficiently clear which steps you took and in which order. Most importantly, **it should be clear how you did your train/validation/test split**. You should prevent leakage, and give us enough information so we can assess leakage.

4. Which model(s) you chose to implement

Which baseline model did you choose? Which other model(s) did you implement? Why did you choose these models? You don't need to explain the details of these models, but you may need to explain enough about the model to understand your choice of model. If you use a novel/unusual model you may need to explain how it works at an intuitive level. You do not need to show the math/algorithms behind standard models. Here you should also discuss hyperparameter optimization and regularization methods.

5. How your model performed (Results)

You should evaluate your model and baseline model with appropriate metrics. You should also observe whether there is overfitting. Additionally, we expect a proper comparison between your model and the baseline. There needs to be some measure of variance. Standard error of mean or Confidence Interval is recommended. See the relevant lecture for advice.

6. What did you do to make your project professional?

Here you may present readme and dependency management, but also containerization, issue tracking, branch management, pre-commit hooks, github actions, etc.

7. Which extras you did that may boost your grade

What things did you do that are exceptionally good work? These can be the things listed in the Grading Specifications document, but you can also propose other things that you did that are really good. I recommend reserving enough time for this if you have many things to show here.

8. Insights from the project as related to the problem at hand

Here you can discuss what you've learned about the problem. This should be something other than "which model is best". Instead you may describe e.g. similarity between classes, high levels of noise in the data, (non-)linearity of the problem, mislabelled samples, sensitivity to preprocessing etc. You can discuss many insights, but they should be related to the problem.